In reality, black holes, unlike the reassembling egg, are compulsory in certain common situations; but only in the late 1960s, when the evidence that they *are* compulsory became overwhelming, did most physicists begin to take black holes seriously. In the next three chapters I shall describe how that evidence mounted from the 1930s through the 1960s, and the widespread resistance it met.

This widespread and almost universal twentieth-century resistance to black holes is in marked contrast to the enthusiasm with which black holes were met in the eighteenth-century era of John Michell and Pierre Simon Laplace. Werner Israel, a modern-day physicist at the University of Alberta who has studied the history in depth, has speculated on the reasons for this difference.

"I am sure [that the eighteenth-century acceptance of black holes] was not just a symptom of the revolutionary fervour of the 1790s," Israel writes. "The explanation must be that Laplacian dark stars [black holes] posed no threat to our cherished faith in the permanence and stability of matter. By contrast, twentieth-century black holes are a great threat to that faith."

Michell and Laplace both imagined their dark stars as made from matter with about the same density as water or earth or rock or the Sun, about 1 gram per cubic centimeter. With this density, a star, to be dark (to be contained within its critical circumference), must have a mass about 140 million times greater than the Sun's and a circumference about 3 times larger than the Earth's orbit. Such stars, governed by Newton's laws of physics, might be exotic, but they surely were no threat to any cherished beliefs about nature. If one wanted to see the star, one need only land on a planet near it and look at its light corpuscles as they rose in their orbits, before plummeting back to the star's surface. If one wanted a sample of the material from which the star was made, one need only fly down to the star's surface, scoop some up, and bring it back to Earth for laboratory study. I do not know whether Michell, Laplace, or others of their day speculated about such things, but it is clear that if they did, there was no reason for concern about the laws of nature, about the permanence and stability of matter.

The critical circumference (horizon) of a twentieth-century black hole presents quite a different challenge. At no height above the horizon can one see any emerging light. Anything that falls through the horizon can never